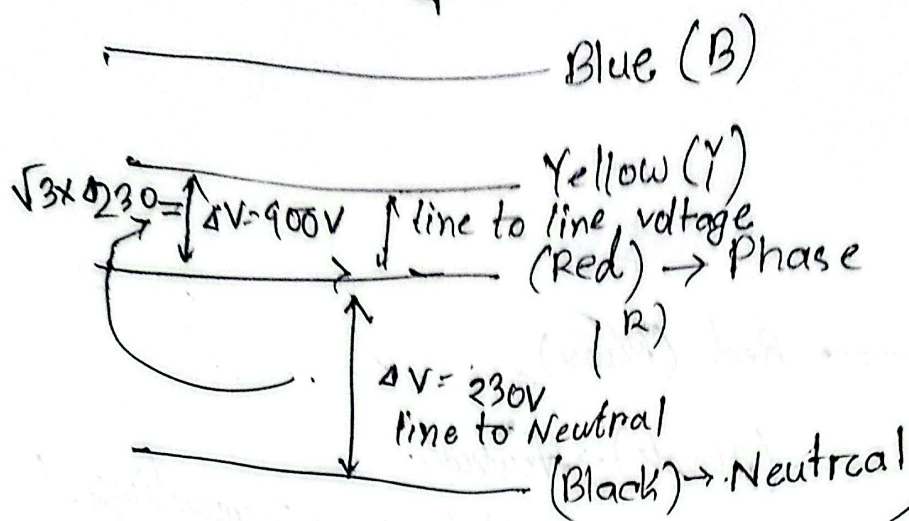
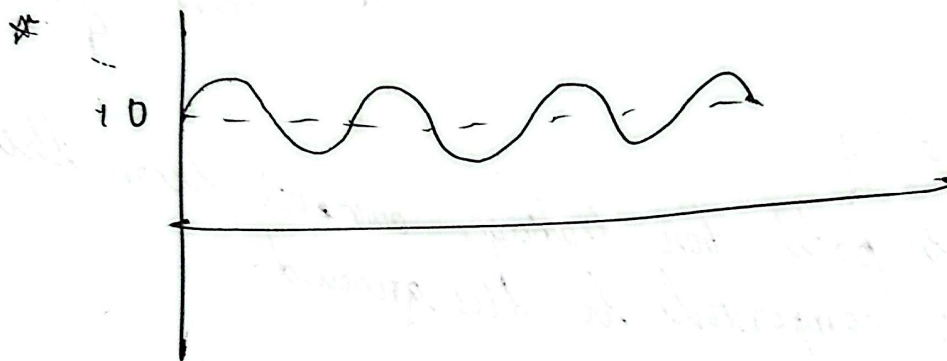


Three phase System:-



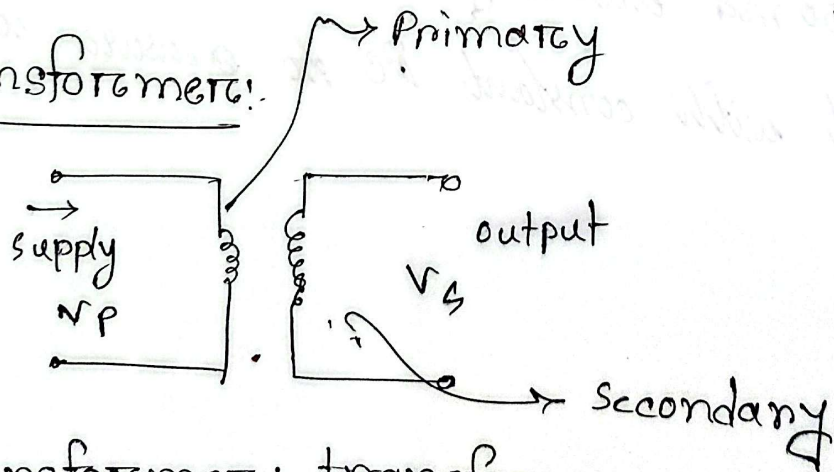
Cu loss is less in three system.
for (High load)

(skin effect)
↓
current has a tendency to be by the surf



$$10 + 100 \sin \omega t$$

Transformer:-



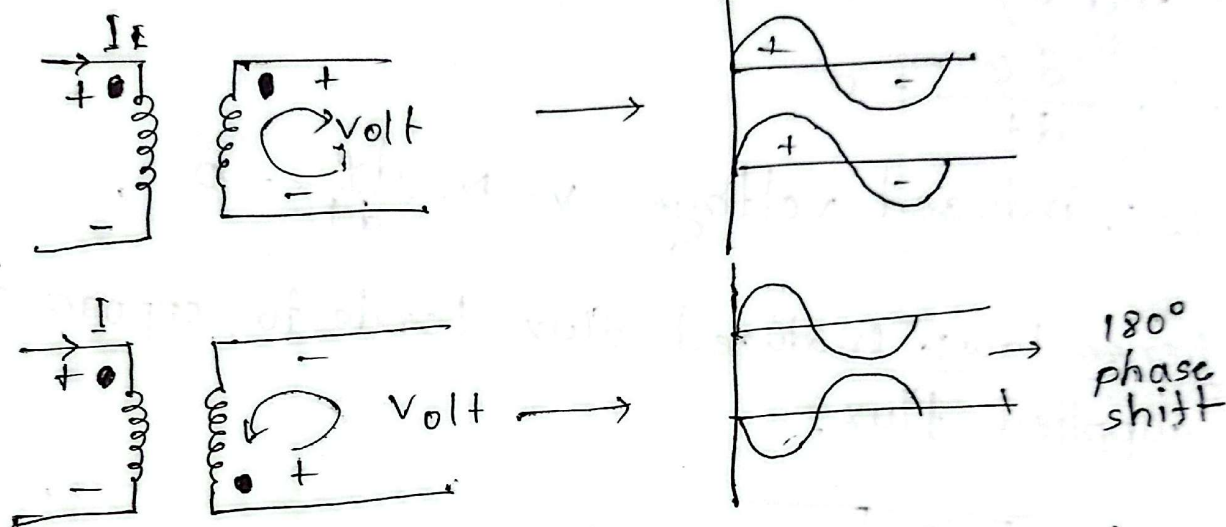
Transformer:- transfers current without any connection.

voltage and current changes but frequency and power remain constant.

for AC, $P = VI \cos \theta$ (θ doesn't change)
 $\downarrow$
 pf

$$V_P I_P = V_S I_S$$

Dot Convention: (Used for Induction).

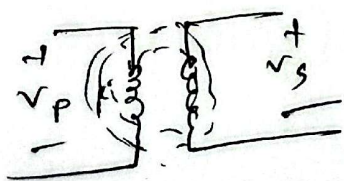


Dot convention is followed for determining the direction of current.

Induced voltage: voltage is induced without any electrical connection.

* Why output level is zero if DC supply is provided in supply?

⇒ DC is not a varying current. As a result it will not obey the Faraday's law of Electro-magnetic induction.



Law of Faraday:-

$$V = -N \cdot \frac{d\phi}{dt} = L \cdot \frac{di}{dt}$$

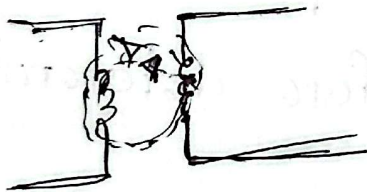
$$\phi = L \cdot i \rightarrow \phi \propto i$$

For electromagnetic induction ϕ and i must be varying. But for dc, $i = \text{constant}$ and $\phi = \text{constant}$.

$$\therefore \frac{d\phi}{dt} = 0$$

$$\therefore \text{induced voltage, } V = N \cdot \frac{d\phi}{dt} = 0$$

Lenz's Law:- Produced flux tends to oppose the supplied flux.

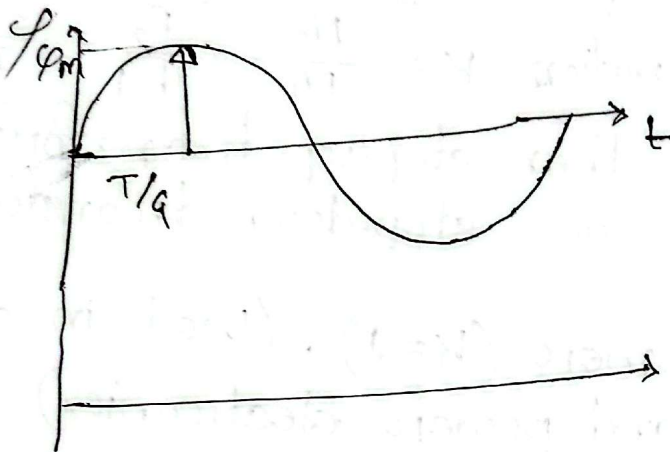


Faraday's law of Electromagnetic induction:-
For development of induced voltage, there must be a relative motion between the magnetic flux and the coil.

Next class:- Transformers in detail.

- Transformer is a static electric device as there is no moving parts or rotational parts.
- That transfers electric energy from one circuit to another without any connection and without changing any frequency.

Induced E.M.F in transformer:-



$$\text{form factor} = \frac{\frac{V_m}{\sqrt{2}}}{\frac{2 V_m}{\pi}} = 1.11$$

Area
 $\phi_m = A \cdot B_m$
 is same for primary and secondary, per turn

Average rate of change of flux, within $T/4$:-

$$\frac{\phi_m}{T/4} = \frac{\phi_m}{1/4f} = 4f\phi_m$$

$$e_{ind} = N \cdot \frac{d\phi}{dt}$$

/total induced emf

Total average rate of change of flux, $= 4fN\phi_m$

$$\begin{aligned} \text{R.M.S value of induced emf} &= \text{Form factor} \times 4fN\phi_m \\ &= 1.11 \times 4fN\phi_m \\ &= 4.44fN\phi_m \\ &= 4.44fNA \cdot B_m \end{aligned}$$

(for both primary and secondary)

Induced emf in Primary, $E_1 = 4f N_1 \phi_m$

Induced emf in Secondary, $E_2 = 4f N_2 \phi_m$

$$\therefore \boxed{\frac{E_2}{E_1} = \frac{N_2}{N_1}}$$

* turns ratio, $a = \frac{1}{k} = \frac{N_1}{N_2} = \frac{E_1}{E_2} = \frac{I_1}{I_2}$

* transformation ratio, $k = \frac{N_2}{N_1} = \frac{E_2}{E_1} = \frac{I_1}{I_2}$

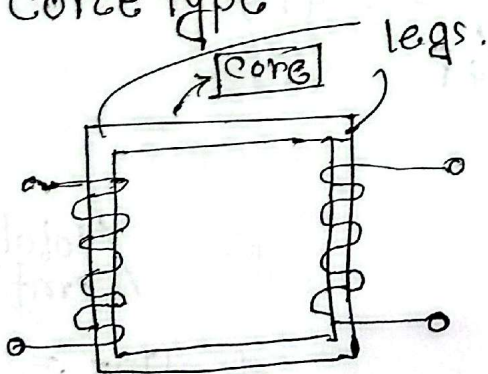
If, $k > 1$, then $a < 1$, then step-up transformer.

If, $k < 1$, $a > 1$, " step-down transformer

→ Isolation transformer ($k=1$). (Used in communication system and power electronics).

Basic types of Transformer (by Construction):

(i) Core type

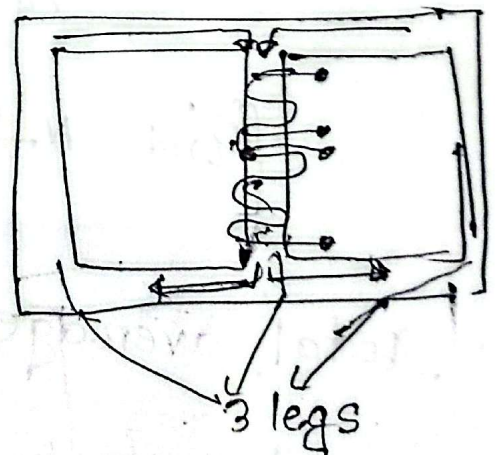


Laminated Steel core,
Used for decreasing Area,

$$\uparrow R = \rho \cdot \frac{L}{A} \downarrow$$

Resistance is considered to be close to ∞ .

(ii) Shell type



Ensures mechanical damage protection.

Core type

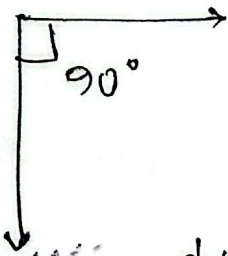
Flux path = 1

Better cooling

Num. of turns can be increased easily

* Efficiency is similar for both types.

* Elementary theory of Ideal transformer.
lagging pf (inductive load).



$$E_2 = -N_2 \frac{d\phi}{dt}$$

$$E_1 = -N_1 \frac{d\phi}{dt}$$

$$= -N_1 \phi_m \omega \cos \omega t$$

$$= N_1 \phi_m \omega \sin(\omega t - 90^\circ)$$

$$\phi = \phi_m \sin \omega t$$

$\therefore E_1, E_2$ lags by 90° from the ~~lags~~ flux.

Shell type

Flux path = 2

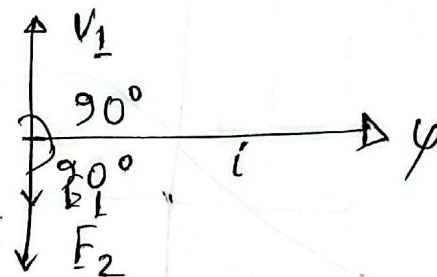
Poor cooling

can't be increased easily.

(losses)

Elementary theory of Ideal transformer.

lagging pf (inductive load).



V_1 = Supply voltage

E_1, E_2 = Induced voltage.

Limits the flow of current.

* Transfer with no load:

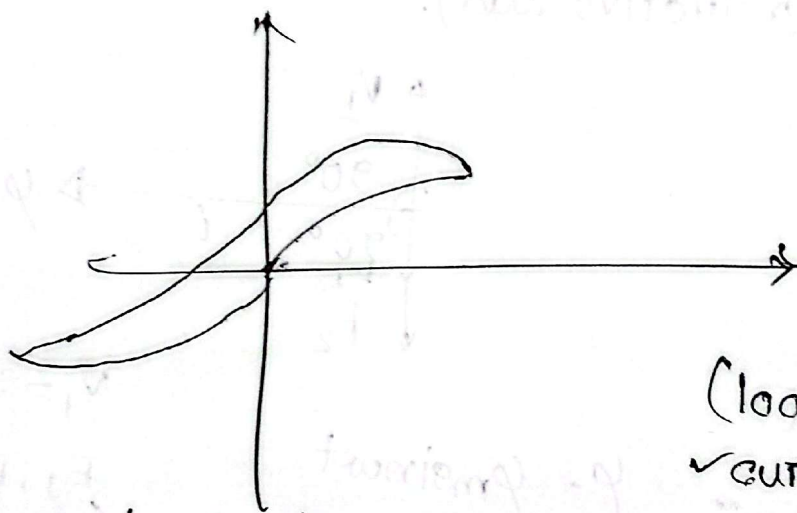
(i) Cu loss $\rightarrow I^2 R$ (Cu loss = 0 for ideal inductor)

$\downarrow$
(Exists in both primary and secondary winding)

(ii) Core loss $\rightarrow$

(i) Eddy current loss ($I_{\text{core}}^2 R_{\text{core}}$)

(ii) Hysteresis loss



Hysteresis loss.

(load বাড়ানো মানে
✓ current বাড়ানো)
× Resistance বাড়ানো

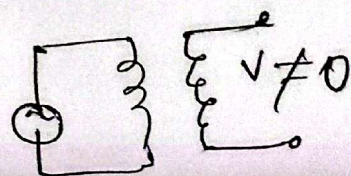
* No load on Secondary side:-

$$\therefore I_s = 0$$

$$\therefore P_s = V_s I_s = 0$$

$\therefore$ Input power should be zero ideally.

But there will be a power loss due to copper loss ($I^2 R$) and core loss in input side. (There will be induced voltage in sec. side but no current).

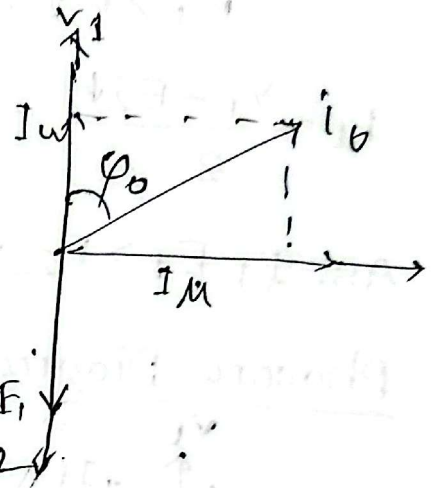


Core loss:-

$I_w \rightarrow$ working/active component (cu loss).
 $I_\mu \rightarrow$ Magnetizing component (Core magnetize loss)
 $\rightarrow$ Always present.

$I_0 \rightarrow$ No load current. (In primary side)

I_w and V_1 in same direction
 as I_w produces due to cu Resi-
 stance and P and I_w remains
 in same direction



I_μ is produced due to
 induction so it lags by $\pi/2$.

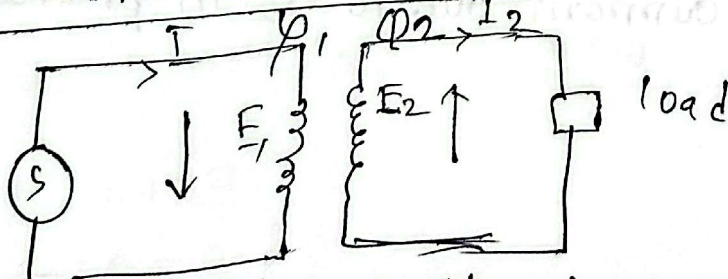
$$I_w = I_0 \cos \phi_0$$

$$I_\mu = I_0 \sin \phi_0$$

$\phi_0 \rightarrow$ No load power factor

$$\therefore I_0^2 = I_w^2 + I_\mu^2 \quad \left[\text{It is always present no matter load exists or not} \right]$$

Transformer on load:-



the direction of flux is opposite in primary side with respect to secondary side.

$\phi_1 =$ flux in primary side

$\phi_2 =$ flux in secondary side.

In no load condition,

$$\text{Resultant } \phi = \phi_1 - 0 \quad (E_1)$$

In on load,

$$\text{Resultant } \phi = \phi_1 - \phi_2 \quad (E_2)$$

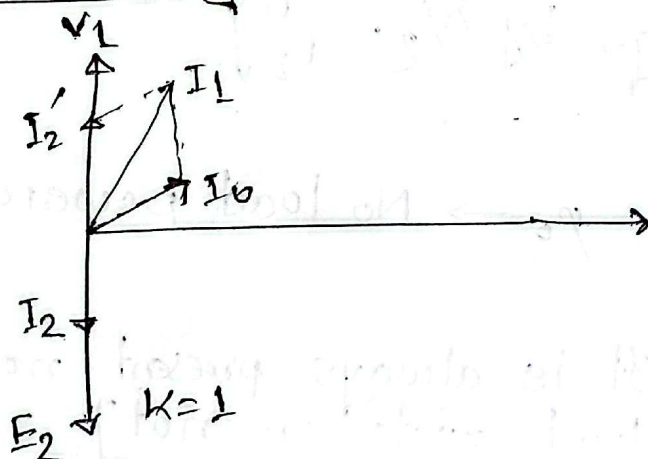
$$E_1 > E_2$$

$$I_1 = \frac{V_1 - E_2}{Z}$$

AC, Bulb, fan
inductive type
load

As, $E_1 > E_2 \therefore I_1 > I_0$

Phasor Diagram:- (For unity power factor).



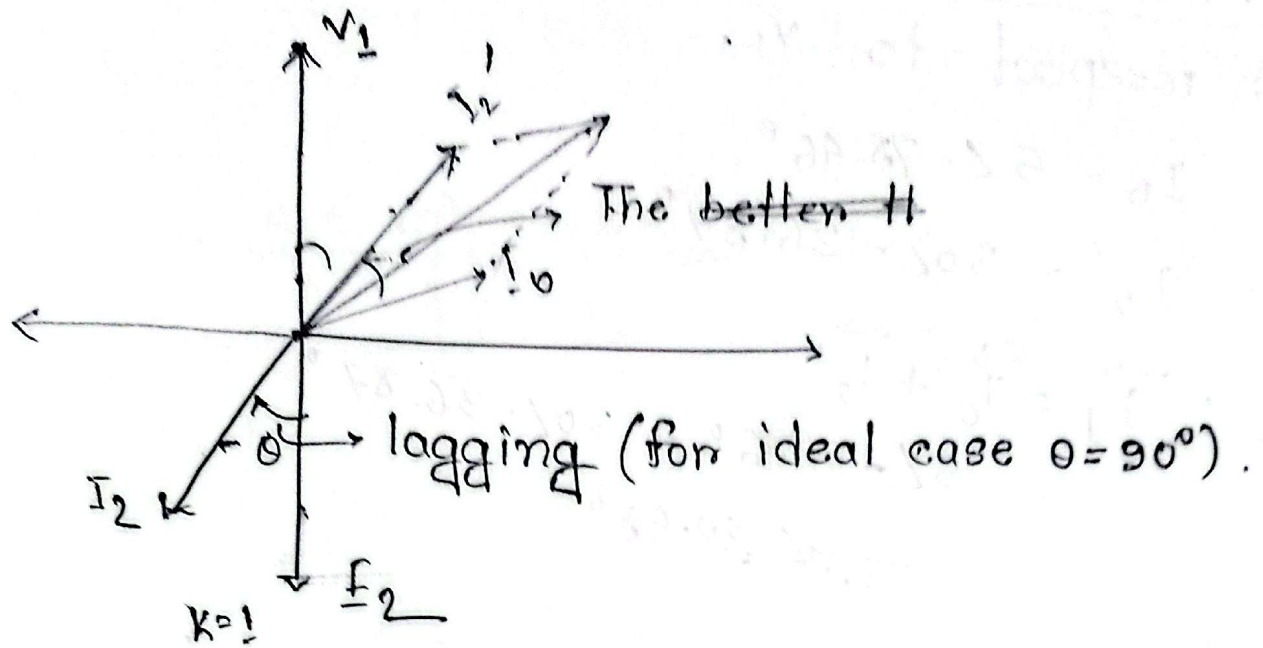
$$K = \frac{I_1}{I_2'}$$

→ current due to I_2 in primary

$$\Rightarrow I_2' = K I_2$$

$$I_1 = I_2' + I_0$$

Forc lagging case:-



p.f কমে গেলে bill বেশি হয়।

Example:- 32.12, 32.11

$$\frac{32.10}{k} = \frac{110}{440} > \frac{1}{9}$$

$$I_0 = 5A$$

$$\cos \theta_0 = 0.2$$

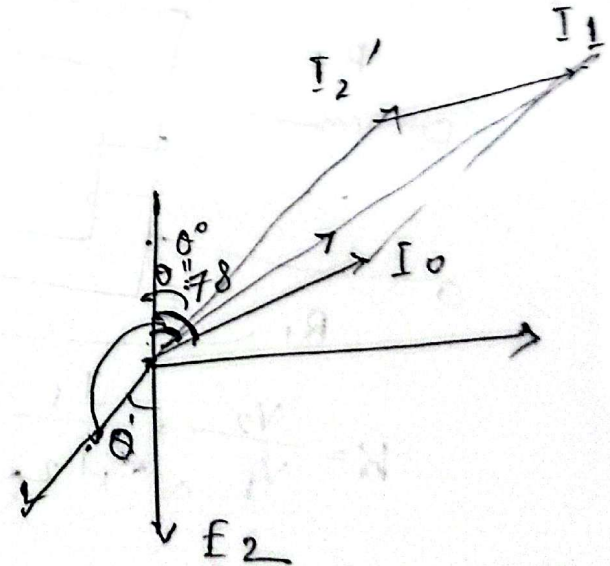
$$\Rightarrow \theta_0 = \cos^{-1}(0.2) = 78^\circ 30'$$

$$I_2 = 120$$

$$\cos \theta = 0.8$$

$$\Rightarrow \theta = 36.87^\circ$$

$$I_2' = k I_2 = 30A$$



Angle between, θ_0 and I_2'
 $= 78^\circ 30' - 36.87'$

$$\therefore I_1 = \sqrt{I_0^2 + I_2'^2 + 2 I_0 I_2' \cos \theta}$$

Alternative method:-

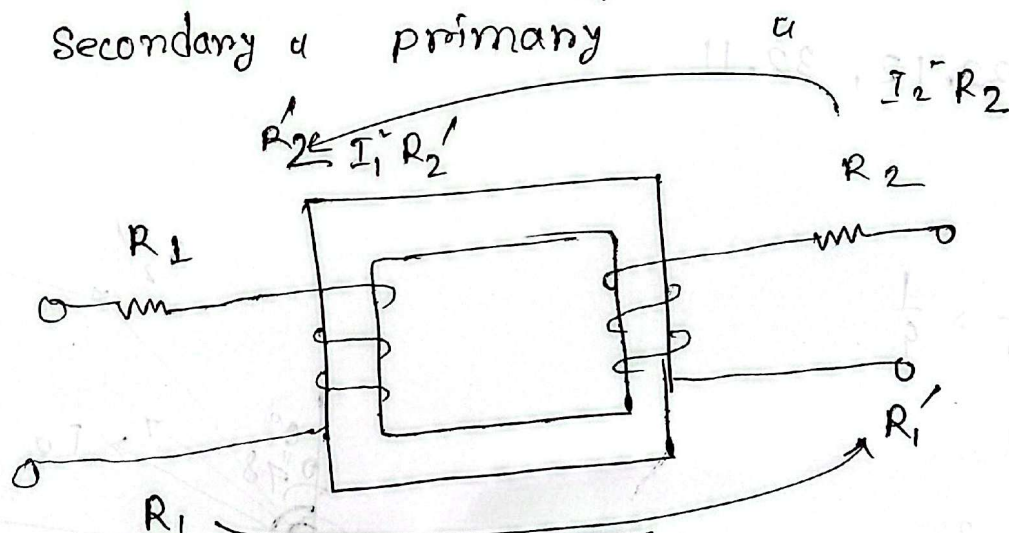
with respect to V_1 ,

$$I_0 = 5 \angle -78.46^\circ$$

$$I_2' = 30 \angle -36.87^\circ$$

$$\begin{aligned} \vec{I}_1 &= \vec{I}_0 + \vec{I}_2' \\ &= 5 \angle -78.46^\circ + 30 \angle -36.87^\circ \\ &= 33.9 \angle -42.48^\circ \end{aligned}$$

* Equivalent circuit of Transformer:-
primary to secondary referred



$$K = \frac{V_2}{V_1} = \frac{I_1}{I_2} = \sqrt{\frac{R_2}{R_1}}$$

Similarly, voltage and current can also be referred

$$I_1' R_2' = I_2 R_2$$

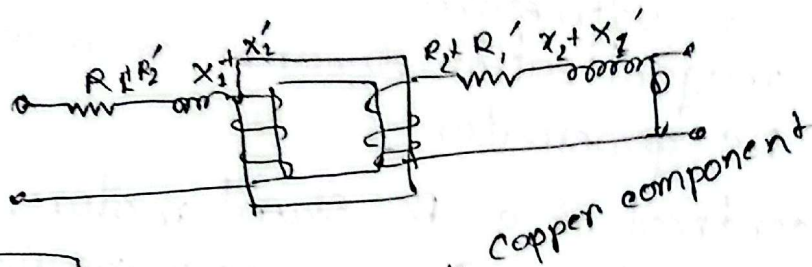
$$\Rightarrow R_2' = \left(\frac{I_2}{I_1} \right) \times R_2$$

$$\Rightarrow R_2' = \frac{R_2}{K^2}$$

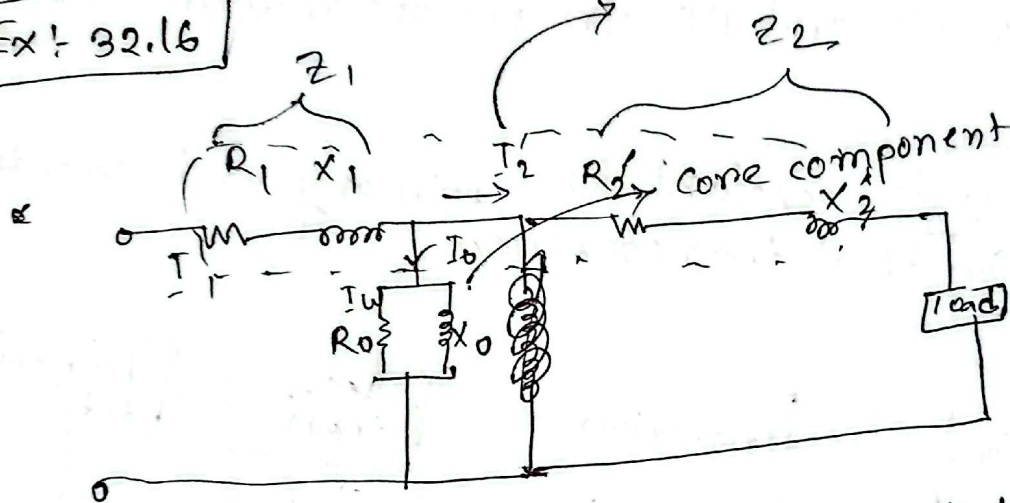
$$R_1' = K^2 \times R_1$$

* Magnetic leakage:

coils contain both reactance and resistance part.

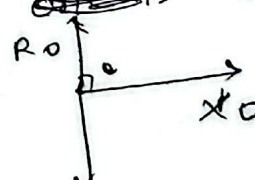


Ex: 32.16



Load is not internal parameter of transformer

I_0 splits because R_0 and X_0 are parallel component.
They are parallel because R_0 is lagging X_0 by 90° .

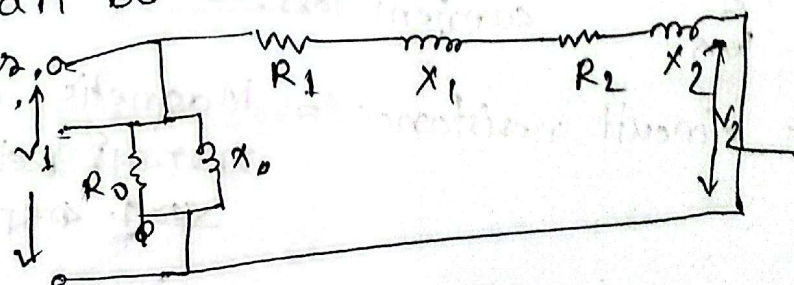


I_0 is very small than I_2 .

$$I_1 \approx I_0 + I_2'$$

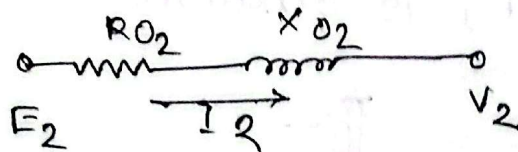
$$I_1 \approx I_2'$$

so, it can be considered that Z_1 and Z_2 are in series.



Phasor Diagram:-

(i) lagging p.f. (Inductive load): Referred into secondary.



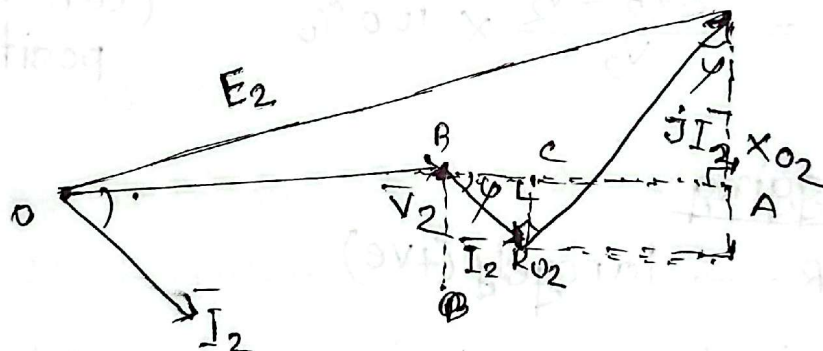
E_2 = No load voltage. or $0V_2$.

V_2 = Full load voltage.

$$\bar{E}_2 = \underbrace{\bar{I}_2 R_{02} + j\bar{I}_2 X_{02}}_{\text{voltage drop}} + \bar{V}_2 \quad \xrightarrow{\text{Needs to be remembered}}$$

* অক্ষ বরাবর যেটা নেওয়া হবে যেটা Reference.

i) Needs to determine the direction of I first.



OA = No load voltage.

OB = full load voltage

AB = Voltage drop.

$$= 0V_2 - V_2$$

$$BC = I_2 R_{02} \cos \phi$$

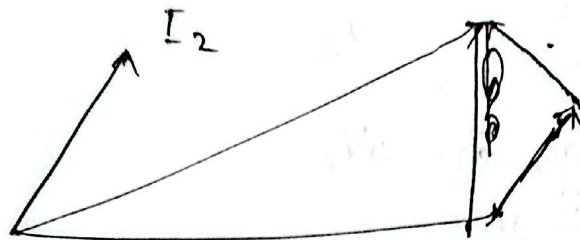
$$AC = I_2 X_{02} \sin \phi$$

$$\therefore |\text{Voltage drop}| = I_2 X_{02} \sin \phi$$

for lagging
 $0V_2$ very large
in comparison
with V_2

(ii) Unity power factor: E_1 moderately longer than V_2

(iii) Leading pf: E_1 is smaller than V_2



Voltage Regulation:-

$$VR_{\text{down}} = \frac{V_{02} - V_2}{V_{02}} \times 100\%$$

$$VR_{\text{up}} = \frac{V_{02} - V_2}{V_2} \times 100\%$$

(can be either positive or negative)

Forc lagging:-

$VR \rightarrow \text{largely (+ve)}$

Forc unity:-

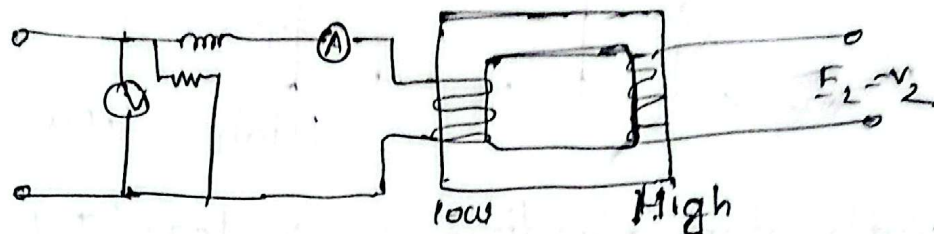
$VR \rightarrow \text{small (+ve)}$

Forc leading:-

$VR \rightarrow (-ve)$

Transformer Test:

(i) Open circuit test (No load test):



Core components and core loss are obtained from this test.

* Why the test is performed in low voltage side?

Rating of transformer 11 kV / 415 V and apparent power 400 kVA

$S = \sqrt{3} V_L I_L$

$\downarrow$ $\downarrow$ $\downarrow$
 20.99 A 556 A

{ For no load test $\rightarrow$ Rated volt supply (Easy)
 " short circuit $\rightarrow$ " current "

If the test was performed in high voltage side, then the rating of voltmeter should be 11 kV and the supply voltage should be high, which is difficult.

Core loss

↑
Wattmeter reading

voltmeter reading

$$P = V I_0 \cos \theta$$

$$\Rightarrow \cos \theta = \frac{P}{V I_0}$$

Ammeter reading

$$\boxed{R_0, X_0}$$

$\downarrow$
 I_w

$\downarrow$
 I_μ

Core component

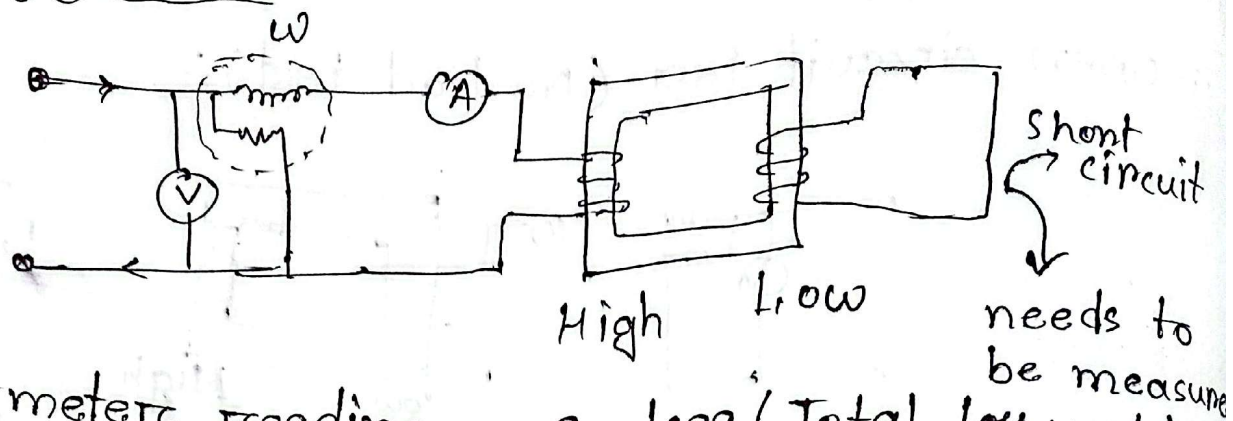
$$I_w = I_0 \cos \theta$$

$$\therefore R_0 = \frac{V_0}{I_w}$$

$$I_\mu = I_0 \sin \theta$$

$$\therefore X_0 = \frac{V_0}{I_\mu}$$

* Short circuit:-



Wattmeter reading $\rightarrow$ Cu loss (Total loss; not in split form)
It provides the total impedance,

Wattmeter measures the active loss (for the resistor only, not the inductive load).

$$W = I_1^2 R_{01}$$

$$\Rightarrow R_{01} = \frac{W}{I_1^2}$$

$$Z = R_{01} + jX_{01}$$

$$\therefore X_{01} = \sqrt{Z_{01}^2 - R_{01}^2}$$